

Title

Effectiveness and Cost-Effectiveness of Digital-Facilitated Self-Management in Inflammatory Bowel Disease: Results from a Pragmatic Randomized Trial

Abstract

While biologic medications improve inflammation markers in inflammatory bowel disease, symptom burden from pain and fatigue often persists. We randomized 780 participants (mean age 43 years, 58% female) to receive digital self-management support (n=391) or standard outpatient care (n=389) over 12 months. Primary disease-specific quality-of-life measures showed modest improvements in both arms, with between-group differences not reaching statistical significance for the pre-specified primary endpoint. However, secondary analyses revealed consistent trends favoring the intervention across symptom domains. The digital-intervention group accumulated 0.016 additional quality-adjusted life years (95% CI 0.002-0.030). Health system costs were £304.66 lower (95% CI -£803.51 to £194.18) in the intervention arm, driven primarily by reduced outpatient visit frequency. Cost-effectiveness analysis indicated the intervention yielded cost savings of approximately £28,633 per additional quality-adjusted life year gained from the healthcare perspective. Results suggest digital symptom-focused self-management offers economic benefits regardless of primary endpoint achievement.